

Victor Espiritu Santiago

AI & ENTERPRISE ARCHITECT · BBVA PERU

Lima, Peru · Email: espiritusantiagovictor@gmail.com · Portfolio: root1v.github.io/portfolio
LinkedIn: linkedin.com/in/victor-espiritu-santiago-89873782 · GitHub: github.com/Root1V

SUMMARY

AI & Enterprise Architecture lead at BBVA Peru, working at the intersection of business strategy, technology architecture, and innovation — defining how a regulated financial institution adopts agentic AI, GenAI platforms, and cloud-native architecture without compromising security or compliance. 15+ years from software developer to enterprise AI leadership: led teams of up to 50 people across three architecture domains, ran a Holding-funded core banking program for three years, and co-defined the Group's AI Decision Framework in Madrid. Still ships code — maintains a self-hosted lab of production-grade AI infrastructure, fully open source, and has built a transformer LLM from scratch as self-directed study.

CORE SKILLS

ARCHITECTURE	Enterprise Architecture · TOGAF · Cloud-Native · Hybrid Cloud · Microservices · gRPC · Event-Driven · Core Banking
AI / ML	Agentic AI · Agent Architectures · Generative AI · LLM Inference · LLM Internals & Fine-tuning · AIOps
GOVERNANCE	AI Governance & Regulation · AI Security · Build vs. Buy Evaluation · FinOps for AI · Pilot Governance
ENGINEERING	Python · Rust · Java / Spring · .NET · Kubernetes · OpenTelemetry
LEADERSHIP	Technology Strategy · Budget & Capacity Management · Cross-Border Coordination · Technical Mentoring · Public Speaking

EXPERIENCE

AI & Enterprise Architecture Lead — BBVA Peru — 2024 – Present

- Define the bank's AI & Cloud architecture strategy; presented a mixed large/small-model architecture strategy (cost and latency driven) to the CIO.
- 2025: international assignment (PDI Program) with BBVA Group's holding AI Architecture team in Madrid — part of the team that co-defined v1 of the Group-wide AI Decision Framework (no-code / low-code / build per use case), presented to the Group's top leadership, and a layered, multi-vendor agentic reference architecture.
- Defined the local AI pilot governance model with Strategy & Control involved from day one; translated Peru's new AI regulation into design rules (customer-facing agents disclose they are AI, generated content is labeled).

Lead Core Banking & Cloud Architect — BBVA Peru — 2019 – 2024

- Product Owner, then Sponsor, of the Horizon Alpha Core Banking transformation (case study below); led up to ~50 people (20 internal, 30 external across four consulting firms) spanning Core, APX, and Host architecture and mobilized 100+ more across country and Holding teams, managing capacity budget on an FTE-cost basis.
- Drove the legacy-to-NextGen transaction migration from 5% (2018) to 71% (2026): transaction volume grew 5x while cost grew 2x (2020–2024).
- Represented engineering on the bank's Innovation Leadership team.

Backend Architecture Manager — BBVA Peru — 2018 – 2019

- Promoted to manage the bank's global backend architectures (ASO, APX, Mainframe) and its private cloud platform (Ether), leading internal and external teams. Taught architecture standards at BBVA's internal Tech University (2018–2020).

APX & Private Cloud Architect — BBVA Peru — 2017 – 2018

- Directed the evolution of the transactional architecture (APX) and the bank's private cloud (Ether); 2018 international assignment to Madrid HQ (see Global Assignments).

Platform as a Service (PaaS) Architect — BBVA Peru — 2016 – 2017

- Led the evolution of the bank's services/API platform (ASO), defined microservices and event-driven standards, and built online and streaming inference so channels consume ML models in real time, not only in batch.

IT Architect / SOA Analyst — Equifax — 2015 – 2016

- Designed architectures for critical projects (Self-Service, Smart SVI, Datamart Financiero) and large-scale data processing pipelines (OSTRA).

Software Technical Lead — CSTI, Softmarkperu, Universidad JFSC — 2010 – 2015

- Led multidisciplinary development teams building systems in Java/Spring, PHP, and .NET for healthcare, telecom, and public-sector clients — the hands-on foundation later architecture decisions drew on.

LEADERSHIP CASE STUDY — HORIZON ALPHA

Role: Product Owner → Sponsor **Program:** Horizon Alpha (Holding-funded, >€1M over 3 years) **Where:** BBVA Peru

When: 2019–2024

BBVA Peru needed to evolve its core banking platform to enable new digital products without disrupting critical operations. I led the strategy for that evolution: 11 of the new Core's 16 reusable capabilities implemented across data, banking architecture, and multichannel layers, so products are built on top instead of rebuilding customer, consent, or signature management each time.

- Enabled the Retail transformation (access, consents, global position, transaction history in the mobile app) and Business banking processes (mass payroll payments, multi-signature approvals, business mobile app).
- Implemented the core functional architecture across the bank's technical stacks — services (ASO), backend (APX), middleware (SEMaaS), batch (LRABatch), frontend (Cells Web) — with BBVA Group's global architecture teams, keeping the system modular and scalable.
- Mobilized 100+ people across the country and the Holding to deploy each Alpha Core release (1,500+ components) in under a quarter with quality automation — the first country in the Group with a zero-impact release.

3x
faster time-to-market: 3–4 quarters → 1–1.5 per product

95%
KPI compliance for rollout and evolution

33%
of the country's strategic initiatives running on the new Core

11/16
reusable core capabilities implemented; ~7 products across Retail and Business

GLOBAL ASSIGNMENTS

2018 · MADRID, SPAIN

Microservices & gRPC architecture

Designed the bank's new containerized microservices architecture built on gRPC, deployable on BBVA's hybrid cloud platform, working directly with the Group's global architecture team.

2025 · MADRID, SPAIN — INTERNATIONAL PDI PROGRAM

Group AI Architecture team

Joined BBVA Group's holding AI Architecture team to help design the Group's agentic AI architecture: a reference architecture and the Decision Framework for choosing the right technology for each use case.

Day to day, coordination with BBVA Group's international architecture teams across Spain, Mexico, Colombia, Argentina, and Turkey.

SPEAKING & RECOGNITION

SEPTEMBER 2026 · LIMA, PERU

C-level panel — Semana Económica

Represented BBVA on a panel for a private club of C-level executives from Peruvian companies, alongside Microsoft and Sonda. Featured in an executive magazine profile ahead of the event.

JUNE 2026 · LIMA, PERU

Investors Peru Event

Spoke on how BBVA brings its AI strategy into the conversation with institutional clients — landing strategy in concrete use cases, not just slides.

SELECTED OPEN-SOURCE PROJECTS — GITHUB.COM/ROOT1V

- **`prometheus-inference-platform`**
  Self-hosted LLM inference gateway (llama.cpp, MLX, vLLM, SGLang) with JWT auth, per-model authorization, OpenTelemetry tracing.
- **`aerarium-agentic-banking`**
  Rust double-entry ledger for agentic commerce — AI agents as first-class account holders, hold-to-capture authorizations.
- **`mercatus-agentic-payments`**
  Framework for autonomous agentic commerce with swappable x402 (USDC) and AP2 (mandate-based) settlement adapters.
- **`veritium-document-intelligence`**
  Agentic document intelligence: hexagonal architecture, bounded ReAct extraction, CEL-rule validation engine, deterministic grounding, pluggable OCR, OpenTelemetry tracing.
- **`synaptum-framework`**
  Durable Python agent runtime — checkpoint/resume, deterministic step identity, zero required dependencies.
- **`aeon-agent-harness`**
  Self-hosted agent harness: durable Temporal execution, typed evidence, policy outside the model, governed memory; interop with LangGraph, CrewAI, OpenAI/Claude Agent SDKs.

CERTIFICATIONS & TRAINING

- **CS50: Artificial Intelligence with Python**
  Harvard University
- **Deep Learning**
  MIT
- **Deep Learning Specialization**
  DeepLearning.AI · Coursera
- **Leadership Program (6 months)**
  BBVA Campus, 2024
- **CS221: AI — Principles & Techniques**
  Stanford University
- **LLM Agents (MOOC)**
  UC Berkeley
- **TOGAF — Enterprise Architecture**
  New Horizons
- **Project Management (PMBOK)**
  PM Criteria